

Two-Dimensional (2D) Nanomaterials: Science, Processing, and Applications (ChE 494)

(Fall 2019; Department of Chemical Engineering, University of Illinois at Chicago)

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Class Schedule: Tuesday and Thursday, 9:30 AM - 10:45 AM, Lincoln Hall

Course Objectives: The goal of this course is to prepare students with foundational understanding of the advanced two-dimensional (2D) nanomaterials including graphene, tools required to characterize such atomically-thin materials, and their potential applications. This course will help students to answer the following questions:

- What a 2D material is and how are they different from their bulk 3D counterpart?
- How to produce these atomically-thin 2D materials and their quantum heterostructures?
- What are their unique and superlative properties and how to characterize them?
- What are the range of potential applications of the 2D materials?

Detailed Course Contents:

Science of 2D Materials (3 Weeks): Low-Dimensional Materials; Layered Materials; 2D Materials; 3D vs. 2D Materials; Modifications of Materials Properties: Doping in Bulk and 2D Materials; Fundamentals of Graphene, Hexagonal Boron Nitride, and Transition Metal Dichalcogenides; and Science of van der Waals 2D Heterostructures. (This part will focus on physics and chemistry of 2D materials, their unique structure and intriguing properties).

Processing of 2D Materials (5 Weeks): Synthesis Techniques for Advanced 2D Materials (Graphene, Hexagonal Boron Nitride, and Transition Metal Dichalcogenides) and their van der Waals bound 2D Heterostructures. (This part will focus on both top-down and bottom-up strategies for the synthesis of 2D materials and understanding the ensuing growth mechanisms).

Characterization and Application of 2D Materials (5 Weeks): Analysis of 2D Materials by Raman Spectroscopy, X-Ray Photoelectron Spectroscopy, Scanning Electron Microscopy, Transmission Electron Microscopy, and Scanning Probe Microscopy. Important Applications of 2D Materials including Nanoelectronics (Field Effect Transistors), Energy Conversion (Photovoltaic Solar Cells), and Energy Storage (Batteries). (This part will focus on basic working principles of advanced characterization tools, detailed analysis of 2D materials, and their potential applications).

2D Lab Work (2 Weeks): Synthesis of Graphene on Metal Films *via* Chemical Vapor Deposition (CVD), Its Chemical Transfer, and Raman Spectroscopic and AFM Characterization.

Grading Policy: Quiz: 40%; Homework Assignments: 10%; Mid-term Review Article: 25%; End-term Project Presentations: 25%.

Textbooks and References:

- (1) 2D Materials: Properties and Devices (P. Avouris, T. F. Heinz, and T. Low)
- (2) Raman Spectroscopy in Graphene Related Systems (A. Jorio, M. S. Dresselhaus, R. Saito, G. Dresselhaus)
- (3) Journal Articles: Nature, Science, Nano Letters, ACS Nano, JACS, PRL, PRB, and More